Publicly Available Satellite and Soil Data Sources for Agricultural Evaluation

Portals, cloud platforms, thematic products and ancillary datasets | October 2026

The table lists free or freely registrable sources suitable for crop monitoring, yield estimation and soil-property mapping. Cloud platforms (Earth Engine, Planetary Computer, Copernicus Data Space) are the most efficient route for time-series work because they avoid downloading raw scenes; agency portals remain the authoritative archives. Ancillary soil and climate datasets are included because they supply the calibration data and covariates that satellite-based soil-health models require.

Source / portal	Operator	What you get	Access
Copernicus Data Space Ecosystem (dataspace.copernicus.eu)	ESA / European Commission	Sentinel-1 SAR (GRD, SLC), Sentinel-2 L1C/L2A, Sentinel-3 OLCI/SLSTR, Sentinel-5P; Copernicus Contributing Missions; Sentinel Hub and openEO APIs, JupyterLab	Free registration; full, open and immediate
Copernicus Land Monitoring Service (land.copernicus.eu)	European Environment Agency / JRC	Global Land products (LAI, fAPAR, NDVI, soil water index, land cover), CORINE, High-Resolution Layers, European crop-type layers	Free
USGS EarthExplorer / Landsat Collection 2 (earthexplorer.usgs.gov)	USGS	Landsat 1-9 archive since 1972 incl. Level-2 surface reflectance and temperature; ARD tiles; also hosts MODIS, ASTER, SRTM and declassified imagery	Free registration
NASA Earthdata / LP DAAC / NSIDC (earthdata.nasa.gov)	NASA	MODIS and VIIRS vegetation indices, LAI/fAPAR, LST, ET (MOD16); Harmonized Landsat Sentinel-2 (HLS); ECOSTRESS; EMIT hyperspectral; SMAP soil moisture; GPM precipitation	Free registration (Earthdata Login)
Google Earth Engine (earthengine.google.com)	Google	Multi-petabyte catalogue (Sentinel-1/2/3/5P, Landsat, MODIS, HLS, SMAP, SoilGrids, ERA5, Cropland Data Layer, WorldCereal, etc.) with server-side processing in JavaScript/Python	Free for research, education and non-profit use; commercial licence otherwise
Microsoft Planetary Computer (planetarycomputer.microsoft.com)	Microsoft	STAC catalogue of Sentinel-1/2, Landsat, MODIS, HLS, NAIP, ESA WorldCover, gNATSGO, Daymet; Python-first Hub	Free; API key for some assets
Registry of Open Data on AWS (registry.opendata.aws)	Amazon / data providers	Cloud-optimised Sentinel-2 (COGs), Sentinel-1, Landsat, NAIP, MODIS, NOAA GOES, NASA HLS	Free (requester-pays for some buckets)
EO Browser / Sentinel Hub (apps.sentinel-hub.com/eo-browser)	Sinergise / Planet	Browser visualisation and index calculation for Sentinel, Landsat, MODIS; custom scripts	Free tier
Theia Land Data Centre (theia.cnes.fr)	CNES / French partners	Sentinel-2 L2A processed with MAJA, SPOT World Heritage, snow and land-cover products	Free registration
EnMAP, PRISMA and DESIS hyperspectral portals	DLR (EnMAP, DESIS), ASI (PRISMA)	30 m hyperspectral imagery (400-2500 nm) for soil mineralogy / SOC and canopy chemistry; tasking requests for new acquisitions	Free for scientific use after registration and proposal
Alaska Satellite Facility (asf.alaska.edu)	NASA / University of Alaska	Sentinel-1 SAR with on-demand RTC and InSAR processing; ALOS PALSAR; NISAR	Free (Earthdata login)

Source / portal	Operator	What you get	Access
JAXA G-Portal	JAXA	AMSR2 soil moisture, GCOM-C vegetation products, ALOS-2 (restricted)	Free registration
USDA NASS CropScape / Cropland Data Layer (nassgeodata.gmu.edu/CropScape)	USDA NASS	Annual 30 m crop-type map for the conterminous USA since 2008; crop frequency layers; VegScape NDVI condition	Free
ESA WorldCereal (esa-worldcereal.org)	ESA / VITO consortium	Global 10 m seasonal cropland, maize and cereal maps plus irrigation layers; open processing system	Free
FAO WaPOR (data.apps.fao.org/wapor)	FAO	Evapotranspiration, biomass production and water productivity for Africa and the Near East at 20-300 m, dekadal	Free
GEOGLAM Crop Monitor (cropmonitor.org)	GEO / NASA Harvest / UMD	Monthly global crop-condition assessments and underlying NDVI, precipitation and soil-moisture anomalies	Free
SoilGrids (soilgrids.org)	ISRIC - World Soil Information	Global 250 m maps of SOC, total N, pH, CEC, bulk density, texture and coarse fragments at six depths with 5th/95th percentile uncertainty; WCS/WMS and REST API	Free (CC BY 4.0)
ESDAC - LUCAS Topsoil (esdac.jrc.ec.europa.eu)	European Commission JRC	Harmonised lab data (SOC, N, P, K, pH, CEC, texture, VNIR spectra) for ~20,000+ EU sites (2009, 2015, 2018); derived European soil-property maps	Free registration
USDA NRCS Web Soil Survey / gSSURGO / gNATSGO	USDA NRCS	Detailed US soil survey polygons and attribute tables (pH, CEC, organic matter, texture, hydrologic group)	Free
Copernicus Climate Data Store - ERA5 / ERA5-Land	ECMWF	Hourly reanalysis weather and soil moisture/temperature grids used as covariates and crop-model forcing	Free registration
Open software for processing	ESA, OSGeo, community	SNAP (Sentinel toolboxes), QGIS, GDAL, rasterio, xarray, scikit-learn, geemap, Orfeo ToolBox, FORCE, Sen2Cor	Free and open source

Practical notes

- Prefer Level-2 surface-reflectance products (Sentinel-2 L2A, Landsat Collection 2 L2, HLS) so that multi-date analyses are physically consistent.
- Sentinel-2 revisit is 5 days at the equator (2-3 days at mid-latitudes with overlapping orbits); Landsat 8+9 together give 8 days; combine with Sentinel-1 for cloud-free continuity.
- Hyperspectral missions (EnMAP, PRISMA, EMIT) are not wall-to-wall; check archive coverage or submit tasking requests early.
- For soil models, pair imagery with LUCAS (EU), NRCS/NCSS (USA) or national soil archives; harmonise depth intervals and laboratory methods before training.
- Check licences: Copernicus, Landsat, MODIS, SoilGrids and CDL are fully open; Earth Engine is free only for non-commercial use; PlanetScope requires a licence (education and research programmes exist).